KICD SENIOR SCHOOL CURRICULUM • GRADE 10

INFORMATION AND COMMUNICATION TECHNOLOGY

Learner's Notes

🗐 **THESE ARE STUDY NOTES**

Based on the KICD Grade 10 ICT Curriculum Design, covering:

- Strand 1.0 – ICT and Society
- Strand 2.0 – Productivity Tools
- Strand 3.0 – Internet and Web Technologies

Compiled for learners aged 14–16 years, Grade 10, Senior School.
These notes are a study companion aligned to the KICD Grade 10 ICT Curriculum Design (June 2024) and are intended to support, not replace, classroom teaching.

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How to Use These Notes

Welcome to your Grade 10 Information and Communication Technology (ICT) Notes! This book has been written specially for you, following the official KICD Senior School Curriculum Design. It is organised into **three strands** and **nine topics**, each broken down into simple, easy-to-follow sections.

As you read through each topic, look out for these special boxes:

Box	What it means
💡 Idea Box	A key fact, definition or tip you should remember.
🔎 Case Study	A real-life or everyday example that shows how the topic applies in Kenya and the world.
🤔 Reflect	An open question with no single "correct" answer – think, discuss with a classmate, or write your own opinion.
🗝 Key Vocabulary	Important new words and their meanings.
✏ Worksheet / Activity	A hands-on or written activity to practise what you have learnt.
📋 End of Topic Questions	Questions to test your understanding after each topic.

All model answers to the Worksheets, Reflect questions and End of Topic Questions are found in the **Answer Key** chapter at the back of this book. Try to answer the questions on your own first before checking the answer key!

1.0

ICT AND SOCIETY

In this strand you will learn what ICT is, where it is used in everyday life, and how operating systems help us manage digital devices.

STRAND 1.0: ICT AND SOCIETY • SUB-STRAND 1.1

Topic 1: Introduction to ICT

By the end of this topic, you should be able to:

- Identify technologies used in communication.
- Explain the components of ICT infrastructure in an organisation.
- Use ICT to interact with information on a topical issue.
- Appreciate the importance of ICT in society.

1.1 What is ICT?

Every day you interact with technology – a mobile phone, a radio, a television, or a computer. All these are examples of Information and Communication Technology (ICT) at work. To understand ICT fully, we must first understand a few basic terms.

🗝 Key Vocabulary

Term	Meaning
Data	Raw, unorganised facts and figures (e.g. numbers, words, images) that have not yet been processed.
Information	Data that has been processed, organised or arranged so that it is useful and meaningful.
Process	A series of actions performed on data to convert it into information.
Communication	The act of sending and receiving messages between two or more parties.
Technology	The application of scientific knowledge to create tools, machines and systems that solve practical problems.
Information Technology (IT)	The use of computers and software to manage information.
Information and Communication Technology (ICT)	The combination of computing and telecommunication technologies used to create, store, process, transmit and receive data and information.

💡 Idea Box

Think of ICT as IT (computers and software) *plus* Communication (telephones, networks, the internet). It is this combination that allows us to send a photo taken in Nairobi and have it viewed instantly in Mombasa or New York!

1.2 Technologies Used in Communication

Communication technologies are devices and systems that allow people to exchange information over distances. They can be grouped as follows:

Category	Examples
Broadcast media	Radio, Television
Telecommunication devices	Mobile phones, landline telephones, fax machines
Computing devices	Desktop computers, laptops, tablets
Networking devices	Routers, modems, switches, satellites
Internet-based platforms	Email, social media, video conferencing apps

🔎 Case Study: M-PESA and Communication Technology

In 2007, Safaricom launched M-PESA, a mobile money transfer service. M-PESA relies entirely on ICT – it uses mobile phone networks (communication technology) and computer servers (information technology) to allow millions of Kenyans to send money, pay bills and buy goods instantly, even in areas without a bank. Today, M-PESA processes millions of transactions daily and has become a model studied around the world.

1.3 Components of ICT Infrastructure in an Organisation

ICT infrastructure refers to all the physical and non-physical resources that support the flow, storage, processing and analysis of data in an organisation, such as a school, hospital, or company.

Component	Description
Hardware	Physical devices such as computers, servers, printers, scanners, and networking equipment.
Software	Programs that run on hardware, e.g. operating systems, word processors, accounting systems.
Networks	Systems that connect devices together, e.g. Local Area Networks (LAN), Wi-Fi, the Internet.
Data/Databases	Organised collections of information stored for use by the organisation.
People (ICT personnel and users)	System administrators, technicians, and everyday users who operate the systems.
Procedures	Rules and policies that guide how ICT resources are used, e.g. an ICT usage policy.

🧠 **Reflect**

Think about your school office or a shop near your home. Which of the ICT infrastructure components listed above have you seen there? What do you think would happen if one of these components (e.g. the network) suddenly failed?

1.4 Using ICT to Interact With Information

ICT devices allow us to search for, receive, and send information on any topic of interest – whether it is checking today's weather, reading news about a county government project, or researching for a class assignment. Common ways of interacting with information include:

- Searching for information using a search engine (e.g. Google, Bing).
- Reading digital newspapers or e-books.
- Sending and receiving information through email or messaging apps.
- Watching educational videos online.
- Listening to podcasts or radio streams.

1.5 Importance of ICT in Society

- **Education:** Enables e-learning, research and access to digital libraries.
- **Business:** Supports e-commerce, mobile banking, and efficient record-keeping.
- **Health:** Enables telemedicine, digital patient records, and health information campaigns.
- **Governance:** Enables e-government services such as e-Citizen in Kenya.
- **Social life:** Enables people to stay connected through social media and instant messaging.

✏ Worksheet 1.1: My ICT Day

List all the ICT devices and services you interacted with yesterday, from morning to evening (e.g. radio alarm, mobile phone, television). For each one, write down whether it is mainly used for communication, information processing, or both.

📋 End of Topic 1 Questions

1. Define the terms "data" and "information", giving one example of each.
2. Distinguish between Information Technology (IT) and Information and Communication Technology (ICT).
3. List and briefly explain any four components of ICT infrastructure in an organisation.
4. Describe two ways in which ICT has improved communication in Kenya.
5. Explain the importance of ICT in any two sectors of society (e.g. education, health, business).

STRAND 1.0: ICT AND SOCIETY • SUB-STRAND 1.2

Topic 2: Application Areas of ICT

By the end of this topic, you should be able to:

- Identify application areas of ICT in society.
- Relate the application areas of ICT to solving community problems.
- Appreciate the role of ICT in society.

2.1 Where is ICT Applied?

ICT is applied in almost every sector of the economy and daily life. Below are some of the major application areas:

Sector	Application of ICT
Education	E-learning platforms, digital libraries, online research, computer-based testing
Business and Commerce	E-commerce (online shops), mobile money, point-of-sale (POS) systems, inventory management
Health	Electronic health records, telemedicine, health information systems, medical imaging
Agriculture	Weather forecasting apps, market price information (e.g. via SMS), precision farming, drone monitoring
Government	e-Citizen services, digital land registries, online tax filing (iTax/KRA)
Banking and Finance	Mobile banking, ATMs, online banking, mobile money transfer
Transport	GPS navigation, ride-hailing apps, digital traffic management, e-ticketing
Entertainment	Streaming services, online gaming, digital music and video

💡 **Idea Box**

An "application area" simply means a field or sector where a technology is put to practical use to solve a real problem.

🔎 **Case Study: Kenya's iTax System**

The Kenya Revenue Authority (KRA) introduced iTax, an online platform that allows citizens and businesses to register for tax obligations, file tax returns, and make payments without physically visiting a KRA office. This has reduced tax filing time, cut corruption opportunities, and increased tax compliance across the country. It is a clear example of how ICT is applied in government to solve the community problem of slow, paper-based tax administration.

2.2 Using ICT to Solve Community Problems

Many everyday community problems can be solved or eased using ICT. For example:

- Farmers use SMS-based services to get up-to-date market prices for their produce, avoiding exploitation by middlemen.
- Community health volunteers use mobile apps to track and report disease outbreaks quickly.
- Local administrators use social media and WhatsApp groups to pass emergency alerts (e.g. flood warnings) to residents.
- Students in remote areas access online learning materials, reducing the effect of teacher shortages.

🤔 **Reflect**

Think of a problem in your local community (for example, poor access to information, slow service delivery, or insecurity). How could an ICT-based solution help address this problem? What challenges might arise in implementing it (e.g. cost, internet access, digital skills)?

2.3 Challenges Posed by ICT in Society

While ICT brings many benefits, it also presents challenges such as:

- Cybercrime and online fraud.
- Digital divide – unequal access to technology between urban and rural areas, or rich and poor.
- Loss of jobs due to automation.
- Health issues from excessive screen use (eye strain, poor posture).

- Spread of misinformation and fake news.

✏ Worksheet 2.1: ICT in My Community

Visit (or recall a visit to) a nearby office, supermarket, or hospital. In the table below, record two ICT applications you observed and the community problem each one helps solve.

Place visited	ICT application observed	Problem it solves

📋 End of Topic 2 Questions

1. List any five application areas of ICT in society.
2. Explain how ICT can be used to solve a problem in agriculture.
3. Describe two challenges that ICT has brought to society.
4. "ICT has made government services more efficient." Discuss this statement using an example from Kenya.

Topic 3: Operating Systems

By the end of this topic, you should be able to:

- Explain the importance of operating systems in digital devices.
- Identify types of operating systems used in digital devices.
- Use an operating system to organise files and folders.
- Implement security control measures on files and folders.
- Appreciate the use of operating systems in a digital device.

3.1 What is an Operating System?

An **Operating System (OS)** is the main software that manages computer hardware and software resources, and provides common services for computer programs. It acts as an interface between the user and the computer hardware.

💡 **Idea Box**

Without an operating system, a computer is just a collection of electronic parts that cannot understand your commands. The OS is the "manager" that allows you to open apps, save files, and connect a printer.

3.2 Importance of Operating Systems

- Manages computer hardware resources (memory, processor, storage devices).
- Provides a user interface (Graphical User Interface or Command Line Interface) for interacting with the computer.
- Manages files and folders, allowing users to organise their work.
- Runs and manages application programs.
- Provides security features such as passwords and user accounts.
- Manages communication between hardware devices, e.g. printer, keyboard, mouse.

3.3 Types of Operating Systems

Operating System	Common devices
Microsoft Windows	Desktop computers, laptops
macOS	Apple computers (iMac, MacBook)
Linux (e.g. Ubuntu)	Servers, developer computers, some desktops
Android	Smartphones, tablets
iOS	iPhones, iPads

3.4 Types of Operating System Interfaces

- **Graphical User Interface (GUI):** Uses icons, windows, and a mouse pointer, e.g. Windows desktop.
- **Command Line Interface (CLI):** Requires the user to type text commands, e.g. Linux terminal.
- **Menu-driven Interface:** Presents the user with a list of options to choose from, common on ATMs and some dedicated devices.

🔎 **Case Study: Android's Dominance in Kenya**

According to statistics from mobile analytics platforms, Android is by far the most widely used mobile operating system in Kenya, running on the majority of smartphones due to its affordability and wide manufacturer support (Samsung, Tecno, Infinix, Itel). This shows how the choice of operating system affects accessibility of ICT to ordinary citizens.

3.5 Organising Files and Folders

The operating system provides a **File Explorer** (or Finder on macOS) which allows you to navigate, view and manage your files and folders. Basic file management tasks include:

Task	Description
Creating	Making a new file or folder
Renaming	Changing the name of a file or folder
Moving	Transferring a file/folder from one location to another
Copying	Duplicating a file/folder to another location while keeping the original
Searching	Locating a file/folder using its name or properties
Deleting	Removing a file/folder (usually sent to Recycle Bin/Trash first)

💡 **Idea Box**

Good file organisation practice: create a main folder for each subject or project, then use clearly named sub-folders (e.g. "ICT > Term 2 > Assignments") rather than saving everything to the Desktop.

3.6 Securing Files and Folders

Operating systems provide several features to keep files and folders secure:

- **Passwords:** Restrict access to the device or specific files.
- **File permissions:** Control who can read, write, or execute a file.
- **User access controls:** Different user accounts with different privileges (e.g. Administrator vs Standard user).
- **File backups:** Copies of files stored safely in case the original is lost or damaged.
- **File organisation techniques:** Proper naming and folder structuring to reduce the risk of losing files.

🤔 **Reflect**

Have you ever lost an important file because it was not backed up, or could not find a file because your folders were disorganised? What lesson did you learn from that experience?

✏ Worksheet 3.1: Practical File Management

On a computer available to you, perform the following tasks and tick when done:

- [ ] Create a folder named "ICT_Grade10" on the Desktop.
- [ ] Inside it, create three sub-folders named "Notes", "Assignments" and "Projects".
- [ ] Create a text file inside the "Notes" folder and rename it "Topic3_Notes".
- [ ] Copy the file into the "Projects" folder.
- [ ] Set a password/lock screen on the computer, if permitted by your teacher.

📋 End of Topic 3 Questions

1. Define the term "operating system" and state two of its functions.
2. Name three types of operating systems and give one device example for each.
3. Distinguish between a Graphical User Interface and a Command Line Interface.
4. Outline three file management tasks that can be performed using a file explorer.
5. Describe two ways of securing files and folders on a computer.

2.0

PRODUCTIVITY TOOLS

In this strand you will learn how to use word processors, presentation software, and desktop publishing tools to create professional documents, slideshows and publications.

Topic 4: Word Processing

By the end of this topic, you should be able to:

- Explain the importance of word processing in document production.
- Select a word processing tool for document production.
- Create a text document using word processing tools.
- Share a document created in a word processing tool.
- Appreciate emerging trends in word processing tools.

4.1 What is Word Processing?

Word processing is the use of a computer and special software (a word processor) to create, edit, format, save, print and share text documents such as letters, reports, and essays.

🗝 **Key Vocabulary**

Common word processing applications include: **Microsoft Word**, **Google Docs**, **LibreOffice Writer**, **Apache OpenOffice Writer**, and **WPS Office Writer**.

4.2 Importance of Word Processing

- Allows easy editing and correction of text without rewriting the whole document.
- Enables professional formatting (fonts, spacing, alignment).
- Supports spell-checking and grammar-checking to reduce errors.
- Allows documents to be saved, duplicated, and shared electronically.
- Saves time and paper compared to handwriting documents.

4.3 Basic Word Processing Tasks

Task	Description
Creating a document	Opening a new blank file to start typing
Saving a document	Storing the file to a location (e.g. hard disk, cloud) with a suitable name
Navigating	Moving around within the document using the mouse, scroll bar, or keyboard shortcuts
Closing	Exiting the document, ensuring all changes are saved
Retrieving	Opening a previously saved document

4.4 Formatting a Document

Formatting makes a document visually appealing and easy to read. Key formatting features include:

- **Character formatting:** font type, size, colour, bold, italics, underline.
- **Paragraph formatting:** alignment (left, right, centre, justify), line spacing, indentation.
- **Page layout:** margins, paper size, orientation, page numbering.
- **Page design elements:** section and page breaks, styles, headers and footers.

💡 **Idea Box**

Editing tools like the **thesaurus** (finds synonyms), **spell checker**, **grammar checker**, **auto-complete** and **auto-correct** help you produce error-free, polished documents quickly.

4.5 Advanced Features

Feature	Use
Table of contents	Automatically lists headings and their page numbers

Table of figures	Lists all images/diagrams and their page numbers
Headers and footers	Repeated text/graphics at the top or bottom of every page
Hyperlinks	Clickable links to other parts of a document, a file, or a website
Cross referencing	Linking to another part of the same document, e.g. "see Figure 3"
Tables	Organise data into rows and columns
Mail merge	Combining a template document with a data source to produce personalised copies (e.g. invitation letters)
Graphics	Symbols, shapes, images, and text wrapping around them
Track changes and comments	Tools for collaborating with others by showing edits and leaving notes

🔎 Case Study: Mail Merge for School Communication

A school administrator needs to send personalised fee balance letters to 500 parents. Instead of typing each letter separately, she creates one letter template in Microsoft Word and links it to a spreadsheet containing each learner's name and fee balance. Using **Mail Merge**, Word automatically generates 500 personalised letters in a few minutes, saving hours of work.

4.6 Sharing a Document

A completed document can be shared in various ways: printing a hard copy, exporting to PDF, attaching to an email, or uploading to a cloud storage service (e.g. Google Drive, OneDrive) and sharing a link.

🤔 Reflect

Which method of sharing a document do you think is most secure – email, cloud link, or printed hard copy? Give a reason for your answer.

4.7 Emerging Trends in Word Processing

- Cloud-based collaboration allowing multiple people to edit a document at the same time in real time (e.g. Google Docs).
- Artificial Intelligence (AI) features such as smart writing suggestions, automatic summarisation, and voice typing.
- Integration with other productivity tools, e.g. inserting live spreadsheet charts into a document.

✏ Worksheet 4.1: Create a Formatted Report

Using a word processor available to you, create a one-page report about "The Importance of ICT in My School" that includes: a heading formatted in bold and 16pt font, at least one bulleted list, a table with three rows, a header with your name, and a page number in the footer. Save the file as "MyICTReport" and share it with your teacher via email or upload.

📋 End of Topic 4 Questions

1. Define word processing and name three word processing applications.
2. Explain any three formatting features used in a word processor.
3. Describe the mail merge feature and give one practical example of its use.
4. Explain three ways a completed word processing document can be shared.
5. Discuss two emerging trends in word processing productivity tools.

Topic 5: Presentation Software

By the end of this topic, you should be able to:

- Explain the importance of presentation productivity tools.
- Select a presentation productivity tool for creating a presentation.
- Create an interactive presentation for a target audience.
- Deliver a slide presentation effectively on a topical area.
- Appreciate emerging trends in presentation productivity tools.

5.1 What is Presentation Software?

Presentation software is used to design slideshows that combine text, images, audio, video and animation to communicate ideas to an audience. Examples include **Microsoft PowerPoint**, **Google Slides**, **LibreOffice Impress**, **Keynote** and **Canva**.

💡 **Idea Box**

A good presentation is not just "text on a slide" – it uses visuals, minimal words per slide, and a logical flow to keep the audience engaged.

5.2 Importance of Presentation Software

- Helps communicate complex ideas visually and concisely.
- Engages the audience through multimedia and interactivity.
- Provides structure to a speech or lesson.
- Can be reused, shared and edited easily.

5.3 Creating and Editing Slides

A presentation is built up slide by slide. Each slide can include:

- Text boxes and bullet points.
- Images, shapes and icons.
- Charts and graphs.
- Multimedia elements such as audio and video clips.
- Speaker notes (visible only to the presenter).
- Master slides and templates (for a consistent design across all slides).

5.4 Adding Interactivity

Feature	Purpose
Slide transitions	Visual effects when moving from one slide to the next
Animations	Movement effects applied to text or objects within a slide
Hyperlinks	Clickable text, images or shapes that jump to another slide, file, or website
Presenter view	Shows the presenter their notes, timing, and the next slide while the audience only sees the current slide

🔎 **Case Study: Using Canva for Community Awareness**

A group of Grade 10 learners created a presentation using Canva to educate their community on the dangers of drug and substance abuse. They used interactive elements – embedded video clips, animated statistics, and clickable resource links – to make the presentation more engaging than a plain slideshow. Their presentation was well received at a school open day and shared with the local Chief's office for wider community sensitisation.

5.5 Delivering an Effective Presentation

- Maintain eye contact with the audience rather than reading directly from the slide.
- Speak clearly and at a moderate pace.

- Use the presenter view to keep track of notes and timing.
- Engage the audience with questions or interactive elements.
- Practice beforehand to build confidence.

> 💭 **Reflect**
> Recall a presentation you have watched (in class, church, or online) that you found engaging. What made it effective? What would you have done differently?

5.6 Emerging Trends in Presentation Software

- AI-powered design suggestions that automatically arrange text and images attractively.
- Real-time collaboration allowing several people to build a presentation together online.
- Interactive polls and quizzes embedded directly into slides for live audience feedback.

✏ Worksheet 5.1: Build a Topical Presentation

Create a 5-slide presentation on a topical issue in your community (e.g. environmental conservation, road safety, or teenage mental health). Your presentation must include: a title slide, at least one image, one animation or transition effect, one hyperlink, and speaker notes on at least two slides. Present it to a partner and note their feedback below.

📋 End of Topic 5 Questions

1. Define presentation software and give three examples.
2. Explain the importance of using presentation software to communicate ideas.
3. Describe three features that add interactivity to a presentation.
4. Outline three tips for delivering an effective presentation.
5. Discuss one emerging trend in presentation productivity tools.

Topic 6: Desktop Publishing

By the end of this topic, you should be able to:

- Explain the importance of desktop publishing tools in document publication.
- Select a desktop publishing tool for producing publications.
- Create a publication using desktop publishing tools.
- Share a publication created in a desktop publishing tool.
- Appreciate emerging trends in DTP productivity tools.

6.1 What is Desktop Publishing?

Desktop Publishing (DTP) is the use of a computer and specialised software to design and produce professional-quality publications such as newsletters, brochures, posters, flyers, magazines and business cards.

🗝 **Key Vocabulary**

Category	Examples
Open Source DTP software	Scribus, LibreOffice Draw, Scribble
Non-Open Source (proprietary) DTP software	Adobe InDesign, QuarkXPress, Affinity Publisher, Microsoft Publisher

6.2 Importance of Desktop Publishing

- Produces professional-looking publications without needing a commercial printing press for design work.
- Combines text and graphics precisely on a page layout.
- Allows businesses, schools, and organisations to create their own marketing and information materials cheaply.
- Supports creativity through templates, fonts and design tools.

💡 **Idea Box**

Word processors are mainly for text-heavy documents (e.g. essays), while DTP tools are designed for precise page layout and graphic design (e.g. posters and brochures), giving finer control over how text and images are arranged.

6.3 Creating a Publication

Basic DTP tasks include creating, saving, navigating, closing, and retrieving a publication, then applying design features such as:

- Text frames and columns.
- Image placement and cropping.
- Colour schemes and themes.
- Master pages/templates for consistent design.

🔎 **Case Study: A School Newsletter**

The Journalism Club at a Senior School uses Scribus, a free and open-source DTP tool, to design their termly newsletter, "The Eagle's Eye". They combine student articles, photos from school events, and a colourful masthead to produce a professional-looking eight-page newsletter which is printed and distributed to parents during visiting days. Using an open-source tool means the school does not need to purchase expensive software licenses.

6.4 Integrating and Sharing Publications

DTP tools allow a publication to be integrated with other applications and shared through:

- Hard copy printing.
- Exporting to PDF for easy distribution.
- Email attachment.
- Uploading to a website or cloud storage.
- Applying **mail merge** to personalise publications such as certificates or invitation cards.

🤔 **Reflect**

Look at a poster, flyer, or magazine you have seen recently. What design choices (colours, fonts, images, layout) made it attractive or unattractive to you?

6.5 Emerging Trends in DTP

- Cloud-based DTP tools (e.g. Canva) that allow design and collaboration directly in a web browser.
- AI-assisted layout and design suggestions.
- Templates optimised for both print and digital/social media formats.

✏ Worksheet 6.1: Design a Flyer

Using any DTP tool available to you (e.g. LibreOffice Draw, Canva, Microsoft Publisher), design an A5 flyer advertising a school event of your choice (e.g. a sports day or a talent show). Include a title, at least one image, event details (date, time, venue), and a colour scheme. Export your flyer as a PDF and share it with your class group.

📋 End of Topic 6 Questions

1. Define desktop publishing and give two examples of DTP software.
2. Distinguish between open source and non-open source DTP software, giving one example of each.
3. Explain the importance of desktop publishing tools in document publication.
4. Describe how mail merge can be applied in a desktop publishing tool.
5. State two emerging trends in desktop publishing.

3.0

INTERNET AND WEB TECHNOLOGIES

In this strand you will learn about the internet, digital communication platforms, and how to behave responsibly as a digital citizen.

STRAND 3.0: INTERNET AND WEB TECHNOLOGIES • SUB-STRAND 3.1

Topic 7: The Internet

By the end of this topic, you should be able to:

- Explain the purpose of the internet in society.
- Explore common services offered through the internet.
- Identify the fundamental components required to set up an internet connection.
- Use a search engine to gather relevant information.
- Use an email application to communicate with peers.
- Appreciate the positive impacts of the internet on society.

7.1 What is the Internet?

The **Internet** is a vast global network of interconnected computer networks that communicate using standard protocols, allowing billions of devices worldwide to share information.

🗝 **Key Vocabulary**

Term	Meaning
Intranet	A private network accessible only to an organisation's staff or members
World Wide Web (WWW)	A system of interlinked web pages accessed via the internet
Extranet	A controlled private network that allows access to authorised outside users, e.g. suppliers
Website	A collection of related web pages under one domain name
Webpage	A single document on the World Wide Web
URL (Uniform Resource Locator)	The address used to locate a resource on the internet, e.g. www.kicd.ac.ke
Server	A powerful computer that stores and delivers data/services to other computers (clients)
Web browser	Software used to access and view websites, e.g. Chrome, Firefox, Edge, Safari
Search engine	A tool used to search for information on the internet, e.g. Google, Bing
Internet Service Provider (ISP)	A company that provides access to the internet, e.g. Safaricom, Zuku, Jamii Telecom
Surfing	Browsing through websites on the internet

7.2 Common Services Offered on the Internet

Service	Examples
Communication	Email, video calls, instant messaging
E-commerce	Jumia, Amazon, online banking
Education	Online courses, digital libraries, research
Social media	Facebook, X (Twitter), Instagram, TikTok
Financial services	Mobile banking apps, online payments
Remote work tools	Zoom, Google Meet, Microsoft Teams
Gaming	Online multiplayer games

7.3 Setting Up an Internet Connection

To connect to the internet, you generally need:

- A device (computer, smartphone, tablet).
- An Internet Service Provider (ISP) subscription.

- A connection medium: Wi-Fi router, mobile data/hotspot, ethernet cable, or fibre optic cable.
- A modem/router to convert signals for your device.

💡 **Idea Box**

"Tethering" means sharing your smartphone's mobile data connection with another device (e.g. a laptop) via a personal/portable hotspot, USB cable, or Bluetooth.

7.4 Using Search Engines Effectively

Search engines like Google help you find relevant information quickly. Useful search techniques include:

- **Keyword search:** typing specific important words related to your topic.
- **Search operators:** e.g. using quotation marks " " for an exact phrase.
- **Filters:** narrowing results by date, file type, or region.
- **Boolean operators:** using AND, OR, NOT to refine a search.
- **Autocomplete and related searches:** suggestions that help refine your query.
- **Image and video search:** switching search tabs to find non-text results.

🔎 **Case Study: Bridging the Digital Divide with Public Wi-Fi**

Several county governments in Kenya, including Nairobi and Kisumu, have set up free public Wi-Fi hotspots in town centres and parks. This has allowed residents, including students without home internet access, to search for information, apply for jobs online, and access government e-services at no extra cost, demonstrating the internet's role in promoting equal access to information.

7.5 Using Email

Email (electronic mail) allows users to send and receive messages, documents and files over the internet. Useful email customisations include:

- Account setup and organising folders (Inbox, Sent, Drafts).
- Setting up email filters to automatically sort incoming mail.
- Configuring notifications and a personal signature.
- Adjusting security settings (e.g. two-factor authentication).
- Setting up backup and storage options.

🤔 **Reflect**

Why do you think it is important to have a professional-sounding email address (e.g. firstname.lastname@email.com) rather than a casual nickname, especially when applying for opportunities such as internships or college admissions?

7.6 Positive Impacts of the Internet on Society

- Improved access to information and education.
- Enhanced communication across distances.
- Boosted economic activities through e-commerce.
- Enabled remote work and flexible learning.
- Improved delivery of government and health services.

✏ Worksheet 7.1: Search and Email Practice

1. Using a search engine, find three reliable sources of information on "renewable energy in Kenya". Write their titles and web addresses below.

   ____
   ____
   ____

2. Compose and send an email to a classmate summarising what you found. Copy your teacher (CC) on the email.

📋 End of Topic 7 Questions

1. Define the term "Internet" and distinguish it from the "World Wide Web".
2. List any four common services offered through the internet.
3. State three components required to set up an internet connection.
4. Explain two techniques for effective use of a search engine.
5. Discuss two positive impacts of the internet on society.

Topic 8: Digital Communication

By the end of this topic, you should be able to:

- Explain the importance of digital communication in society.
- Identify online platforms used to support digital communication.
- Use digital online platforms to communicate.
- Use online forums to collaborate and share content.
- Appreciate the impact of Artificial Intelligence (AI) in digital communication platforms.

8.1 What is Digital Communication?

Digital communication refers to the exchange of information using digital devices and platforms, such as email, chat apps, video conferencing, and social media, rather than traditional paper-based or face-to-face methods only.

8.2 Importance of Digital Communication

- Enables instant communication regardless of distance.
- Supports remote learning and remote work.
- Facilitates collaboration on shared documents and projects.
- Reduces the cost of communication compared to travel or postal mail.

8.3 Digital Communication Platforms

Category	Examples
Email	Gmail, Outlook, Yahoo Mail
Instant messaging	WhatsApp, Telegram, Messenger
Video conferencing	Zoom, Google Meet, Microsoft Teams
Collaboration platforms	Slack, Microsoft Teams
Learning management systems	Google Classroom, Moodle
Social media	Facebook, X (Twitter), Instagram, TikTok

💡 **Idea Box**

Good digital communication platforms usually offer: user profiles, customisable interfaces, accessibility features, search functionality, privacy and security controls, integration with other services, and support/help resources.

8.4 Digital Collaboration Tools

Type	Examples	Use
Cloud-based document platforms	Google Docs, Zoho Docs	Editing documents together in real time
File-sharing services	Dropbox, Google Drive, OneDrive	Storing and sharing files online
Project management platforms	Asana, Jira	Planning, assigning and tracking tasks in a project

🔎 **Case Study: Group Project Using Google Docs**

A group of Grade 10 learners working on a group history project used Google Docs to write their report together. Even though the members were in different homes during the school holiday, they could all type into the same document at the same time, see each other's edits live, and leave comments for one another – without needing to meet physically or email the file back and forth.

8.5 Streaming Media Platforms

Video and audio streaming platforms such as YouTube, Netflix, Spotify, Prime Video, and Facebook Live allow content to be delivered live or on-demand over the internet. These can also be used to create **podcasts** – audio programs that learners can use to share learning experiences or opinions on topical issues.

🤔 **Reflect**

Which digital communication platform do you use most often to talk to friends or family? What features of that platform make it your preferred choice?

8.6 Impact of Artificial Intelligence (AI) on Digital Communication

- AI chatbots provide instant customer support on websites and apps.
- AI-powered translation tools break language barriers in real time.
- AI suggests smart replies and auto-completes messages.
- AI content moderation helps detect and remove harmful or abusive content on platforms.

✏ Worksheet 8.1: Digital Collaboration Task

In groups of three, create a shared document using a cloud-based platform (e.g. Google Docs) on the topic "Benefits and Risks of Social Media for Teenagers". Each member should contribute at least two points, and one member should use the comment feature to give feedback on another's contribution.

📋 End of Topic 8 Questions

1. Define digital communication and give three examples of digital communication platforms.
2. Explain the importance of digital communication in today's society.
3. Distinguish between a file-sharing service and a project management platform, giving one example of each.
4. Describe two ways Artificial Intelligence has impacted digital communication platforms.
5. Explain how online forums can be used to collaborate and share content.

Topic 9: Digital Citizenship

By the end of this topic, you should be able to:

- Explore the importance of being a good digital citizen.
- Identify and observe appropriate netiquette behaviours online.
- Observe healthy practices when using ICT technologies.
- Analyse practices for environmental conservation to mitigate degradation caused by digital technology.
- Appreciate the importance of responsible use of technologies.

9.1 What is Digital Citizenship?

Digital citizenship refers to the responsible, ethical and safe use of technology by individuals in an online environment. A good digital citizen respects others online, protects their own and others' privacy, and uses technology to contribute positively to society.

9.2 Netiquette (Online Etiquette)

Netiquette refers to the accepted rules of polite and responsible behaviour when communicating online. Good netiquette includes:

- Being respectful and polite in comments and messages.
- Avoiding writing in all capital letters (which is seen as "shouting").
- Thinking before posting or sharing content.
- Giving credit to original authors when sharing their work.
- Avoiding spamming or flooding a forum with irrelevant posts.
- Being patient and understanding towards others online.

💡 Idea Box

A helpful rule: before you post anything online, ask yourself – "Would I say this to the person's face?" If the answer is no, reconsider posting it.

9.3 Unethical Behaviours Online

- Cyberbullying – using digital platforms to harass or intimidate others.
- Plagiarism – copying someone else's work and presenting it as your own.
- Spreading misinformation or fake news.
- Hacking or unauthorised access to other people's accounts or data.
- Identity theft and online impersonation.

🔎 Case Study: Fact-Checking Before Sharing

During a period of heavy rains, a message claiming that a certain dam was about to burst spread rapidly on WhatsApp groups in a certain county, causing panic among residents. It was later confirmed by the county government to be false information. This case shows the importance of evaluating information for credibility and verifying facts from official sources before sharing content online, to avoid contributing to public panic or misinformation.

9.4 Digital Footprint and Privacy

A **digital footprint** is the trail of data you leave behind through your online activity (posts, likes, searches, purchases). Maintaining a positive digital footprint means:

- Thinking carefully before posting photos, comments, or personal information.
- Regularly reviewing and adjusting privacy settings on social media and email accounts.
- Being aware that content shared online can be difficult to remove completely, even after deletion.

🤔 Reflect

If a university admissions officer or future employer searched your name online today, what would they find? Does your current digital footprint reflect the person you want to be seen as?

9.5 Healthy Practices When Using ICT

- **Ergonomics:** Sit with good posture, position the screen at eye level, and take regular breaks from screens.
- **Eye care:** Follow the 20-20-20 rule – every 20 minutes, look at something 20 feet away for 20 seconds.
- **Balanced screen time:** Avoid excessive use of digital devices; balance online activities with physical exercise and offline social interaction.
- **Sleep hygiene:** Avoid excessive screen use before bedtime, as blue light can affect sleep quality.

9.6 Environmental Conservation and Digital Technology

The manufacture, use and disposal of digital devices can harm the environment. Good practices to reduce this impact include:

- **Energy efficiency:** switching off devices when not in use, using energy-saving settings.
- **Electronics recycling (e-waste management):** properly disposing of old devices at designated e-waste collection points rather than throwing them in general waste.
- **Digital waste management:** deleting unnecessary files, emails and cloud storage data, since data centres that store this information consume large amounts of electricity.

✏ Worksheet 9.1: Netiquette Role Play

With a partner, role-play a scenario in which one of you posts something rude or inconsiderate on a class WhatsApp group, and the other responds appropriately using good netiquette. Write down the key lines used in your role play below, then swap roles.

📋 End of Topic 9 Questions

1. Define the term "digital citizenship".
2. List any four netiquette behaviours that should be observed in an online environment.
3. Explain the meaning of "digital footprint" and why it is important to manage it carefully.
4. Describe two healthy practices to observe when using ICT devices.
5. Explain two ways in which the use of digital technology can be made more environmentally friendly.

CHAPTER 10

Answer Key: Model Answers to Worksheets, Reflections and End of Topic Questions

Use this chapter to check your work after attempting the questions on your own. For "Reflect" questions, sample points of discussion are given since there is no single correct answer.

Topic 1: Introduction to ICT

Worksheet 1.1: Answers will vary by learner. A strong answer clearly separates devices/services into communication (e.g. mobile phone, radio), information processing (e.g. calculator), or both (e.g. smartphone, computer with internet).

End of Topic Questions:

1. Data is raw, unprocessed facts (e.g. the numbers "70, 65, 80"); Information is processed, meaningful data (e.g. "the average score is 71.7%").
2. IT refers to computers and software used to manage information; ICT includes IT plus communication technologies (networks, telephones, internet) that allow information to be shared between people/devices.
3. Any four of: Hardware (physical devices), Software (programs), Networks (connect devices), Data/Databases (stored information), People (users/technicians), Procedures (usage policies) – with a brief explanation of each.
4. Any two, e.g.: Mobile phones allow instant voice/text communication across distances; the internet/email allows fast exchange of documents and messages; M-PESA allows instant transfer of money using mobile networks.
5. Any two sectors with explanation, e.g. Education (e-learning, digital libraries) and Health (telemedicine, digital records).

Topic 2: Application Areas of ICT

Worksheet 2.1: Answers vary; look for a correctly matched application (e.g. POS machine in a supermarket) to a real problem it solves (e.g. faster, accurate billing and stock tracking).

1. Any five, e.g.: Education, Business/Commerce, Health, Agriculture, Government, Banking, Transport, Entertainment.
2. E.g. SMS-based market price information helps farmers know fair prices for their produce and avoid exploitation by middlemen.
3. Any two, e.g.: Cybercrime/online fraud; digital divide between rural and urban areas; job losses due to automation; health issues from excessive screen use.
4. Learners should discuss with an example such as iTax or e-Citizen, explaining how it has reduced time and improved service delivery, while also noting possible limitations (e.g. requiring internet access).

Topic 3: Operating Systems

Worksheet 3.1: A practical, teacher-assessed task; learners should be able to demonstrate the folder structure and file operations to their teacher.

1. An operating system is software that manages computer hardware and software resources and acts as an interface between the user and the hardware. Functions: managing hardware resources; providing a user interface (any two accepted).
2. Windows – desktop/laptop computers; Android – smartphones/tablets; macOS – Apple computers (any three valid pairs accepted, including Linux – servers, and iOS – iPhones/iPads).
3. A GUI uses icons, windows and a mouse pointer for interaction, while a CLI requires the user to type text commands.
4. Any three, e.g.: creating, renaming, moving, copying, searching, deleting files/folders.
5. Any two, e.g.: setting passwords; applying file permissions/user access controls; keeping backups.

Topic 4: Word Processing

Worksheet 4.1: Teacher-assessed practical task; check for correct formatting, bulleted list, table, header and footer as instructed.

1. Word processing is the use of a computer and software to create, edit, format and print text documents. Examples: Microsoft Word, Google Docs, LibreOffice Writer.
2. Any three, e.g.: font formatting (size, style, colour); paragraph alignment; page layout (margins, page numbers).
3. Mail merge combines a template document with a data source (e.g. a spreadsheet of names) to produce multiple personalised documents automatically, e.g. sending personalised fee balance letters to parents.

4. Printing a hard copy; exporting to PDF and emailing; uploading to cloud storage and sharing a link.
5. Any two, e.g.: cloud-based real-time collaboration; AI-powered writing suggestions and voice typing.

Topic 5: Presentation Software

Worksheet 5.1: Teacher/peer-assessed practical task; check for the five required elements (title slide, image, animation/transition, hyperlink, speaker notes).

1. Presentation software is used to create slideshows combining text, images and multimedia to communicate ideas. Examples: Microsoft PowerPoint, Google Slides, Canva.
2. It helps communicate complex ideas visually, engages the audience, and provides a clear structure for a talk or lesson.
3. Any three, e.g.: slide transitions, animations, hyperlinks, presenter view.
4. Any three, e.g.: maintain eye contact; speak clearly; practise beforehand.
5. Any one, e.g.: AI-powered design suggestions that automatically arrange slide content.

Topic 6: Desktop Publishing

Worksheet 6.1: Teacher-assessed practical task; check the flyer includes a title, image, event details and a coherent colour scheme, exported as PDF.

1. Desktop publishing is the use of a computer and software to design and produce professional publications such as newsletters and flyers. Examples: Scribus, Microsoft Publisher.
2. Open source DTP software is free and its code can be modified (e.g. Scribus); non-open source (proprietary) software is commercially licensed (e.g. Adobe InDesign).
3. It allows individuals and organisations to produce professional-quality publications with precise control over layout, text and graphics, without needing a commercial printer for design.
4. Mail merge in a DTP tool can be used to personalise multiple copies of a publication, e.g. printing certificates with each learner's name automatically inserted.
5. Any two, e.g.: cloud-based DTP tools like Canva; AI-assisted layout suggestions.

Topic 7: The Internet

Worksheet 7.1: Teacher-assessed practical task; check for credible sources listed and a correctly composed and sent email with CC used.

1. The Internet is a global network of interconnected computer networks; the World Wide Web (WWW) is a system of interlinked web pages/documents accessed through the internet using a browser. The internet is the network; the web is a service that runs on it.
2. Any four, e.g.: communication (email), e-commerce, education, social media, financial services.
3. Any three, e.g.: a device, an Internet Service Provider (ISP), a modem/router, a connection medium (Wi-Fi/cable).
4. Any two, e.g.: using specific keywords; using quotation marks for exact phrases; using filters to narrow results.
5. Any two, e.g.: improved access to information/education; boosted e-commerce and economic activity.

Topic 8: Digital Communication

Worksheet 8.1: Teacher-assessed group task; check that all members contributed and the comment feature was used appropriately.

1. Digital communication is the exchange of information using digital devices and platforms. Examples: email, WhatsApp, Zoom.
2. It enables instant communication regardless of distance, supports remote work/learning, and facilitates collaboration.
3. A file-sharing service (e.g. Google Drive) stores and shares files online; a project management platform (e.g. Asana) is used to plan, assign and track project tasks.
4. Any two, e.g.: AI chatbots provide instant support; AI enables real-time translation between languages.
5. Online forums allow users to post questions, share ideas, and respond to one another, enabling collaborative discussion and content sharing on topics of shared interest.

Topic 9: Digital Citizenship

Worksheet 9.1: Teacher/peer-assessed role play; check that the "appropriate" response models respectful, constructive netiquette (e.g. politely correcting behaviour rather than retaliating).

1. Digital citizenship is the responsible, ethical and safe use of technology by individuals in an online environment.
2. Any four, e.g.: being respectful and polite; avoiding writing in all capitals; thinking before posting; giving credit to original authors.
3. A digital footprint is the trail of data left behind by a person's online activities. It is important to manage it because online content can be seen by future employers, colleges, or the wider public and can be difficult to remove once posted.
4. Any two, e.g.: maintaining good posture/ergonomics when using devices; following the 20-20-20 rule for eye care.
5. Any two, e.g.: switching off devices when not in use to save energy; recycling old electronic devices at proper e-waste collection points.